

SECTION A

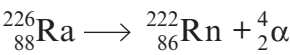
Answer **all** questions.

1. (a) Radon is a radioactive gas and emits alpha particles.
- (i) Explain what is meant by the *activity* of a radioactive source **and** give its unit. [2]
- (ii) The half-life of radon is 3.3×10^5 s. Determine the time it takes for a sample of radon to decay to 20 % of its initial activity. [3]
- (iii) Discuss the extent to which alpha particles are a risk to the human body. [3]



Examiner
only

(b) Radium-226 decays into radon.



Assuming that 98 % of the energy released in this process is converted to the kinetic energy of the alpha particle, use the data below to determine the speed of the alpha particle. [5]

Atomic masses: $^{226}_{88}\text{Ra} = 226.025\,410\,\text{u}$ $^{222}_{86}\text{Rn} = 222.017\,578\,\text{u}$

$^4_2\text{He} = 4.002\,603\,\text{u}$ $m_{\text{electron}} = 0.000\,549\,\text{u}$

$1\,\text{u} = 931\,\text{MeV}$

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